

1. Which of the following is a key characteristic of supervised learning?

- ☒ a) The model learns from labeled data
 - b) The model learns from unlabeled data
 - c) The model learns through trial and error
 - d) The model requires no input data
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2. Which of the following is a common method for handling missing data?

- a) Ignoring missing values
 - ☒ b) Replacing missing values with the mean or median
 - ☒ c) Using machine learning algorithms that ignore missing values
 - d) Both b and c
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3. In decision trees, what is the criterion used to split the data at each node?

- ☒ a) Entropy or Gini index
 - b) Accuracy
 - c) Log-Loss
 - d) Cross-Validation
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4. Which type of machine learning algorithm is typically used for anomaly detection?

- a) Supervised Learning
 - ☒ b) Unsupervised Learning
 - c) Reinforcement Learning
 - d) Semi-supervised Learning
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5. Which algorithm is best suited for a high-dimensional dataset with many features?

- a) K-Nearest Neighbors
 - b) Support Vector Machines with a linear kernel
 - c) Logistic Regression
 - ☒ d) Support Vector Machines with a non-linear kernel
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6. What does the "bias" in a machine learning model refer to?

- ☒ a) The error introduced by the model's inability to fit the training data well
 - b) The error due to the model being too complex
 - c) The difference between the predicted and actual values
 - d) The difference between the model's predictions and the true values for all data points
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7. Which of the following is true for gradient descent?

- ☒ a) It is an optimization technique used to minimize the cost function

- b) It is used to maximize the accuracy of the model
 - c) It helps in reducing the size of the dataset
 - d) It requires only a linear relationship between features and target
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8. Which machine learning algorithm is used for both classification and regression tasks?

- ☒ a) Random Forest
 - b) K-Means
 - c) Support Vector Machine
 - d) K-Nearest Neighbors
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9. What does the "variance" in a machine learning model refer to?

- a) The model's ability to generalize to new data
 - ☒ b) The model's sensitivity to small changes in the training data
 - c) The error due to the model's complexity
 - d) The model's bias towards certain outcomes
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10. Which of the following is a hyperparameter in machine learning?

- a) Model weights
 - ☒ b) Learning rate
 - c) Predicted output
 - d) Training data
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11. Which algorithm is often used for feature selection in high-dimensional datasets?

- ☒ a) Principal Component Analysis (PCA)
 - b) Linear Regression
 - c) Decision Trees
 - d) K-Nearest Neighbors
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12. What is "regularization" used for in machine learning?

- a) To speed up the training process
 - ☒ b) To prevent overfitting by adding a penalty to the model's complexity
 - c) To improve the model's precision
 - d) To increase the size of the training data
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13. Which of the following is a commonly used evaluation metric for regression models?

- a) F1-Score
 - ☒ b) Mean Squared Error (MSE)
 - c) Precision
 - d) Recall
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14. Which of the following is a disadvantage of using K-Nearest Neighbors (KNN)?

- ☒ a) It is sensitive to irrelevant features
 - b) It is difficult to interpret
 - c) It requires a large number of hyperparameters
 - d) It only works for regression problems
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15. Which machine learning algorithm is based on the concept of "neighbors"?

- a) K-Means
 - b) Decision Trees
 - ☒ c) K-Nearest Neighbors
 - d) Support Vector Machines
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